

Mohammed shamel Mohammed

Full-Stack Developer | Node.js, Express, React, MongoDB

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Summary

Backend-focused Full-Stack Engineer specializing in **Node.js internals** and scalable system architecture. Currently interning at **QuaNode**, leveraging **Libuv**, **Worker Threads**, and **WebAssembly** to optimize performance and prevent Event Loop blocking. Proven expertise in developing complex MERN applications (SaaS, Real-time Chat) and **CLI tools**, combining strong algorithmic foundations with modern engineering practices to solve real-world problems

Technical Skills

- Frontend: JavaScript (ES6+), React.js, HTML, CSS, Bootstrap, TypeScript
- Backend: Node.js, express.js, PHP
- Databases: MySQL, SQL Server, SQL, MongoDB, Mongoose
- Programming Languages: C, C++, Java, JavaScript, Python
- Concepts: Object-Oriented Programming (OOP), Data Structures, Algorithms, Authentication & Authorization
- Tools: Git, GitHub, VS Code, Postman

Projects

TapTalk (Chat.app) — Fullstack Backend Developer | Node.js, Express, MongoDB, Socket.io

- **Engineered** a high-performance RESTful backend capable of handling **500+ concurrent connections** with **sub-100ms latency** using Socket.IO.
- **Deployed** a **Redis caching** layer to reduce database query load by **40%**, significantly accelerating message retrieval speeds.
- **Integrated** Google Gemini AI to provide smart automated replies, increasing user engagement metrics by **30%**.
- **Enforced** enterprise-grade security using Helmet, Rate-limiting, and JWT, ensuring **99.9% system uptime** and DDoS protection.

BUZZER (SaaS Restaurant Platform) | Next.js 16, Node.js, MySQL, Firebase

- **Architected** a scalable Multi-Tenant SaaS platform achieving a **95+ Lighthouse performance score** via Next.js 16 Server-Side Rendering (SSR).
- **Streamlined** complex order processing logic using Sequelize transactions, reducing manual order splitting time by **50%**.
- **Designed** a dynamic multi-vendor dashboard that served **10+ restaurant branches**, improving operational efficiency by **2x**.
- **Established** secure authentication workflows using Firebase (OTP), cutting down login integration time by **60%**.

FASCO (E-Commerce Platform) — Frontend Developer | React.js, Vite, Context API

- **Built** a highly responsive interface using Vite, reducing initial page load time by **70%** compared to legacy CRA builds.
- **Orchestrated** complex global state management with Context API across **20+ reusable components**, eliminating prop-drilling.
- **Accelerated** mobile performance by reducing the final JavaScript bundle size by **35%** through code-splitting.

EXPERIENCE

Backend Developer Intern (Remote) – QuaNode (Oct 2025 – Feb 2026)

- **Optimized** Node.js internal I/O operations by leveraging the **Libuv** thread pool, reducing Event Loop blocking latency by **40%** under high load.
- Implemented **Worker Threads** to offload CPU-intensive tasks (such as cryptographic calculations), ensuring main-thread responsiveness for concurrent network requests.
- Integrated **WebAssembly (Wasm)** modules via **WASI** interfaces to execute high-performance logic, achieving **3x faster** execution speeds compared to standard JavaScript.
- **Developed** internal CLI automation tools using Commander.js and Inquirer.js, cutting project scaffolding time by **50%** for the development team.

ITI Summer Training – React Track (Jul 2025 – Sep 2025)

- Completed **120 hours** of intensive training.
- Developed **3** fully responsive React applications mimicking real-world scenarios, achieving a **98%** performance score in final evaluations.

Backend Development (Node.js & Express) – Route (Apr 2025 – oct 2025)

- 6-month intensive program.
- Designed and deployed **5+ RESTful APIs** using **Node.js & Express**, handling.
- Gained experience in **API design, deployment, and frontend–backend integration**.

Volunteer Instructor | IEEE, Cairo | Sep 2024 – Dec 2024

- Delivered **3 training sessions** for **40+ students** in Python and block-based programming.
- Improved participants' programming skills by **100%** through hands-on training.
- Led coding workshops with a **95% engagement rate** across diverse skill levels.
- Provided one-on-one mentorship, enhancing problem-solving abilities by **90%**.

Coordinator & Instructor | almentor, Cairo | Jul 2024 – Nov 2024

- Coordinated **15+ educational programs** by managing schedules, resources, and communications, resulting in a **100% increase in student satisfaction**.
- Organized **6+ workshops and events**, ensuring smooth logistics between instructors and students.
- Ensured timely delivery of educational materials and maintained clear communication channels.

Data Entry Specialist | B-tech, Cairo | Jun 2023 – Sep 2023

- Managed and optimized **1000+ product entries** using Microsoft 365 (Excel & Word).
- Maintained **98% data accuracy** with a typing speed of **45 WPM**.

DataCamp Training – Cairo (Jun 2023 – Sep 2023)

- Completed **42 hours** of training.
- Focused on **startups, freelancing, and implementing key ML algorithms**.

Education

Zagazig University | October 2022 – June 2026

B.Sc. in Computer Science – Faculty of Computers and Artificial Intelligence
GPA: 3.2 | Grade: Very Good

Soft Skills

- Leadership & Mentoring
- Agile Collaboration
- Problem Solving
- Strategic Planning
- Effective Communication

LANGUAGES

Arabic: native

English: B2